Estuti Shukla

Department of Physics
Institute for Gravitation and the Cosmos
The Pennsylvania State University
University Park, PA 16802, USA

estuti@psu.edu
https://eshukla.github.io/

EXPERIENCE

Graduate Student Researcher Aug. 2022 – Present
The Pennsylvania State University — Advisor: Prof. David Radice *University Park, PA*

- Developing scientific machine learning methods to solve fundamental problems in general relativity
- Performing high-resolution simulations of spinning compact binary coalescences using `GR-Athena++` and GPU-capable `AthenaK`, scalable numerical relativity codes for high-performance computing

Visiting Student Researcher Sept. 2021 – Aug. 2022
ICTS–TIFR Bangalore — Advisor: Prof. Prayush Kumar *Bangalore, India*

- Studied properties of dynamical horizons using numerical relativity simulations with the `SpEC`, a pseudo-spectral high-performance computing code for numerical relativity

Research Intern Oct. 2020 – Aug. 2022
Johns Hopkins University — Advisor: Dr. Vishal Baibhav & Prof. Emanuele Berti *Remote*

- Investigated nonlinearities of quasi-normal modes associated with ringdown of binary black hole mergers

Summer Research Intern Jun. – Sept. 2020
University of Massachusetts Dartmouth — Advisor: Prof. Gaurav Khanna & Prof. Scott Field *Remote*

- Developed calculations of Bondi mass and recoil kick velocities from black-hole perturbation theory waveforms
- Contributed to early calculations underlying the open-source `gw_remnant` package; acknowledged in Phys. Rev. D 108, 064048 (2023)

EDUCATION

The Pennsylvania State University State College, USA
Ph.D. in Physics *Aug. 2022 – Present*

- Advisor: Prof. David Radice
- Research: AI4Science, numerical relativity, black hole perturbation theory

Indian Institute of Science Education and Research Kolkata Kolkata, India
B.S.–M.S. in Physics — GPA (major): 9.03/10.00 *2016 – 2021*

- Master's Thesis: *Static Stars in Cosmological Backgrounds*, advisor Prof. Rajesh Nayak
- DST Inspire Fellowship (Govt. of India)

RESEARCH INTERESTS

Scientific machine learning, numerical relativity, high performance computing, and gravitational wave physics.

PUBLICATIONS

[1] **E. Shukla**, D. Radice, R. Gamba, and S. Bernuzzi, "Gauge Fixing with Neural Networks," arXiv (2026).

[2] **E. Shukla**, A. Suriyanarayanan, D. Radice, R. Maulik, and S. Bernuzzi, "Puncture Evolution with Physics-Informed Neural Networks,", submitted to *PRX Intelligence*, arXiv:2610.06776 (2026).

[3] R. Gamba, D. Chiaramello, **E. Shukla**, and S. Albanesi, "Spin the black circle II: tidal heating and torquing of a rotating black hole by a test mass on generic orbits," accepted in *Phys. Rev. D*, arXiv:2603.28982 (2026).

[4] **E. Shukla**, A. Rashti, R. Gamba, D. Radice, and K. Chandra, "`GR-Athena++` simulations of spinning binary black hole mergers," *Class. Quant. Grav.* **43**, 065019 (2026).

[5] R. Perna, O. Gottlieb, **E. Shukla**, and D. Radice, "Connecting GRBs from binary neutron star mergers to nuclear properties of neutron stars," *Phys. Rev. D* **111**, 063015 (2025).

[6] M. H.-Y. Cheung, V. Baibhav, E. Berti, V. Cardoso, G. Carullo, R. Cotesta, W. Del Pozzo, F. Duque, T. Helfer, **E. Shukla**, and K. W.-K. Wong, "Nonlinear effects in black hole ringdown," *Phys. Rev. Lett.* **130**, 081401 (2023).

TALKS & CONFERENCES

Invited Talks

Strong Gravity Seminar, Penn State, PA — Sept. 2026
AstroChat Talk, UC Berkeley, CA — July 2026
Physics Colloquium, UMass Dartmouth, MA *(virtual)* — Sept. 2020

Contributed Talks

ICDS Symposium, Penn State, PA *(Poster)* — Oct. 2026
AstroAI Workshop 2026, Harvard University, MA — Jun. 2026
26th Eastern Gravity Meeting, Cornell University, Ithaca, NY — May 2026
AI for Gravitational Waves Workshop, CERN, Geneva *(Poster)* — May 2026
35th Midwest Relativity Meeting, U. Guelph, Canada — Nov. 2025
Scientific Machine Learning for GW Astronomy, ICERM, RI *(Poster)* — Jun. 2025
IGC Alumni Event, Penn State, PA *(Poster)* — May 2025
APS Global Physics Summit 2025, Anaheim, CA — Mar. 2025

Schools & Workshops

IAIFI Summer School, UMass Boston, MA *(virtual)* — Aug. 2026
Deep Learning for Science School, Berkeley, CA — Jul. 2026
AthenaK Summer School, Penn State, PA — Jul. 2026
Simulating Extreme Spacetimes with SpEC and SpECTRE, ICERM, RI — Aug. 2024
DTP TALENT Summer School on Nuclear Astrophysics, Trento, Italy — Jul. 2024
N3AS Summer School in Multi-messenger Astrophysics, UC Santa Cruz, CA — Jul. 2023
ICTS School on Physics of Early Universe, ICTS-TIFR, India — Jan. 2022
Finesse Interferometer Modelling Hackathon, IUCAA, India *(winner)* — Mar. 2020
National Science (Vijyoshi) Camp, DST, IISER Kolkata, India — Dec. 2016

AWARDS & HONORS

APS DGRAV Student Travel Award, APS Global Summit — 2025
- Awarded $600 in travel support, selected among 93 applicants to attend the conference

EUSTIPEN Travel Award, Trento, Italy — 2024
- Awarded $2,500 travel support by FRIB-TA to attend the DTP TALENT Summer School

Homer F. Braddock Scholarship, Penn State University — 2022-23
- Awarded a $5,000 scholarship from Eberly College of Science in recognition of academic excellence

DST INSPIRE Fellowship, Government of India — 2016-21
- Selected amongst top 1% Class XII board performers
- Awarded an amount equivalent to approximately $1,150 annually

TECHNICAL SKILLS

Programming (Advanced): Python, PyTorch, LaTeX, C/C++, Mathematica
Programming (Intermediate): R, MATLAB
Numerical Relativity Codes: AthenaK, GR-Athena++, SpEC
Tools: Git, GNUplot, Origin, Linux, Claude Code

TEACHING AND MENTORSHIP

Teaching Assistant, Penn State University — 2022-24
- PHYS 212 General Physics: Electricity and Magnetism
- PHYS 250 Introductory Physics 1

Teaching Assistant, IISER Kolkata — 2020-21
- General Theory of Relativity & Cosmology (w/ Prof. Rajesh Nayak)
- Intermediate Quantum Mechanics (w/ Prof. Soumitro Banerjee)

Undergraduate Student Mentor
- Patrick A Bush, San Diego State University (graduate student at OSU)

PROFESSIONAL ACTIVITIES

Journal Referee

- Classical and Quantum Gravity

Member

- LIGO-Virgo-KAGRA Consortium (2022-)
- APS DGRAV, FGSA, FIP (2023-)
- Simulating eXtreme Spacetimes Collaboration (2021-22)

OUTREACH AND SERVICE

Lecture on Gravitational Waves, State College Area High School, PA	Apr. 2026
Professional Development Chair, Physics and Astronomy for Women+, Penn State	2024-27
Physics Graduate Student Representative, Penn State	2025-26
Board Member, Physics Graduate Student Association (PGSA), Penn State	2025-26
Delegate for ECoS, Graduate & Professional Student Association (GPSA), Penn State	2025-26
Chair of Academic Affairs Committee, GPSA, Penn State	2025-26
Founder of IKQRAAR, official LGBTQ+ support platform, IISER Kolkata	2021-22
Founder of Trekking & Adventure Club, IISER Kolkata	2020-21
Founding Member of Student Alumni Cell, IISER Kolkata	2019-20
Sponsorship Head of Inquivesta, college science festival, IISER Kolkata	2019-20

REFERENCES

Prof. David Radice
Associate Professor at the Departments of Physics and Astronomy & Astrophysics
Pennsylvania State University
email: dur566@psu.edu
Prof. Romit Maulik
Assistant Professor of Mechanical Engineering
Purdue University
email: rmaulik@purdue.edu
Prof. B.S. Sathyaprakash
Elsbach Professor of Physics and Professor of Astronomy and Astrophysics
Pennsylvania State University
email: bss25@psu.edu
Dr. Rossella Gamba
Postdoc
University of California Berkeley
email: rgamba@berkeley.edu